

ANKIT KUMAR

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EDUCATION

- **Birla Institute of Technology, Mesra** 2026
B.Tech in Electronics and Communication Engineering CGPA: 8.29
- **Narbheram Hansraj English School** 2022
Indian School Certificate (ISC), New Delhi Percentage: 95
- **Hill Top School** 2020
Indian Certificate of Secondary Education (ICSE), New Delhi Percentage: 94.2

EXPERIENCE

- **Samsung Research & Development, Noida** Jan 2026 - Jun 2026
SDE Intern Noida
 - Built a **Flask-ADB** diagnostic utility adding **illegal IMEI detection** at the customer service tier, removing engineering-escalation dependency and cutting diagnosis time to under 60 seconds; in staged rollout to **300+ Samsung service centers** nationwide.
 - Rebuilt an internal file-transfer server (**Django, Nginx**) that hard-failed above 30 MB, implementing **chunked multipart uploads** with client-side segmentation, **async server-side reassembly**, and SQL-backed chunk-state tracking; removed the size cap, enabled **resumable uploads**, deployed across **5 teams serving 200+ users**.
 - Designed a Carrier Aggregation extraction engine using **PyQt6 and Pandas** indexing 500+ config entries across 10+ Qualcomm spec files into a unified searchable interface, cutting engineer lookup time by **95%+** (5+ min to **under 10 sec**).
- **Tata Motors** May 2025 - Jul 2025
SDE Intern Jamshedpur
 - Built a fault-isolated **ETL pipeline** aggregating 165+ Excel reports per 15-minute cycle from 11 edge sensor systems into a central store; engineered per-source **fault isolation** (file locks, network drops, machine-offline) ensuring zero-downtime ingestion.
 - Built a real-time safety-compliance system (**YOLOv10, Python**) processing **5 live CCTV feeds** at **10 FPS** on standard laptop hardware, with automated **helmet-violation detection** and bounding-box audit logging, achieving **94%+ accuracy** on Bay 11 of the production line.

PROJECTS

- **QueryInsight : AI-Powered SQL Chatbot & Visualization Dashboard** May 2025
Tools & technologies: Python, LangChain, Groq API, SQLAlchemy, MySQL, and Plotly/Dash.
 - Engineered a multi-stage **NL-to-SQL** pipeline using **LangChain** and (**Groq API**) with dynamic schema injection via **SQLAlchemy** reflection, plus a three-stage validation layer (extraction, syntax check, schema cross-referencing) with read-only enforcement, and validation layer to eliminate invalid-column hallucinations before execution.
 - Built an auto-selecting visualization layer in **Plotly/Dash** that infers chart type from query-result schema across **7 chart types**, routing aggregations to bar charts and time-series to line charts, with anomaly detection on returned datasets.
- **Distributed Network Monitoring System** Jun 2026
Tools & technologies :Java, Spring Boot, Apache Kafka, MongoDB, ShedLock, REST API, Docker
 - Built a horizontally scalable, **fault-tolerant** distributed system with independent scheduler, worker, and ingestion roles (one **Spring Boot** app via profiles) over partitioned **Apache Kafka** topics, scaling out as **consumer groups**.
 - Guaranteed single-active scheduling via a **MongoDB**-backed distributed lock (**ShedLock**) and **ordered, idempotent** processing through device-ID partition keying; containerized with **Docker**.
- **Predictive Maintenance (RUL Prediction)** Feb 2025
Tools & technologies: Python, TensorFlow, XGBoost, Pandas, Random Forest, Scikit-learn
 - Built a **RUL prediction** system on **NASA C-MAPSS (FD001)**, removing 7 redundant sensors via Pearson correlation with RUL clipping and 10-cycle rolling-mean augmentation, achieving **RMSE 32.48** (R2 0.71) with **SVR** and 4-fold cross-validation.
 - Extended it to a **3-tier risk classifier** (No Risk, Moderate, Critical) using **SVC** and **Random Forest** with stratified splits, reaching **68%** validation accuracy across imbalanced classes.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Data Engineering: TL/ELT Pipelines, Apache Kafka, Data Modeling, SQL/NoSQL

Cloud: AWS S3, Glue, GCP

DevOps: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Nginx, Linux Shell Scripting

Databases: MySQL, MongoDB, BigQuery

BI Visualization: Plotly, Dash, Tableau

AI / ML: LangChain, Groq API, SQLAlchemy, Scikit-learn, XGBoost, SVR, Random Forest, YOLOv10, Plotly/Dash

ACHIEVEMENTS

- **Leetcode Knight** (rating 1900+) Jun 2026
- **Graduate Aptitude Test in Engineering (GATE) in DA** Secured AIR 1388 Feb 2025
- **Joint Secretary, Robolution Robotics Club** Feb 2024